CAIRO: Decoupling Order and Scale for Regression

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Abstract

Many regression tasks care about both the ordering of predictions and their scale. We propose CAIRO (**C**alibrate **A**fter **I**nitial **R**ank **O**rdering), a simple two-stage framework for regression with real-valued data. Stage 1 trains a score function using ranking losses; Stage 2 fits a one-dimensional isotonic regression to map scores back to the scale of the targets. Empirically, CAIRO acts as a robust regression method, matching the performance of tree ensembles on tabular benchmarks.

1 Introduction

Supervised learning models are typically evaluated against metrics that reflect specific deployment goals. We focus on two distinct families of such metrics. The first consists of regression-based metrics, like mean squared error (MSE), which reward predictions that are numerically close to the target value. The second consists of rank-based metrics, which reward models for correctly ordering pairs of examples relative to one another. While these objectives are often treated separately, a growing number of applications require models to succeed at both simultaneously.

In classification settings, this dual requirement is common, for instance, in ad click prediction or medical screening, where models must rank candidates effectively while also outputting calibrated probabilities. However, this need also extends to regression tasks with continuous, real-valued targets. A prime example is insurance pricing: a model must rank policyholders by risk to segment them correctly, but it must also assign precise premiums to ensure profitability to the insurer.

While methodologies for combining calibration and ranking are well-established in classification, they are under-explored for regression with real-valued targets. Consequently, techniques designed for binary or discrete outcomes do not naturally generalize to continuous settings.

In response, we introduce CAIRO (**C**alibrate **A**fter **I**nitial **R**ank **O**rdering), a two-stage framework designed to optimize both ranking and regression performance for real-valued targets. CAIRO learns

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sequentially: Stage 1 minimizes a ranking loss to learn the optimal ordering, and Stage 2 minimizes a pointwise loss to recover the scale. This framework is theoretically grounded; we show that it targets the true conditional expectation at the population level, just as a standard MSE-based learner would.

Additionally, the separation of learning in stages provides a significant advantage over standard regression: because ranking losses are scale-invariant, the representation learned in Stage 1 is inherently shielded from the magnitude of outliers. This allows CAIRO to function as a robust regression method. It is particularly effective in regimes where the data exhibits heavy tails or heteroskedasticity, offering stability where standard MSE minimization often fails.

The remainder of this paper is organized as follows. Section 2 reviews related work on learning-to-rank and score recalibration. Section 3 introduces the CAIRO framework and formalizes its two-stage population objective. Section 3.1 studies Stage 1, including weighted pairwise ranking losses, their pointwise equivalents, and the class of *Optimal-in-Rank-Order* (ORO) objectives that guarantee order-consistency with the conditional mean. Section 3.2 analyzes Stage 2 and proves that monotone calibration via isotonic regression recovers the true regression function when Stage 1 is ORO. Section 4 details empirical estimation and practical algorithms, including smooth surrogates for efficient optimization. Finally, Section 5 presents synthetic and real-data benchmarks demonstrating that CAIRO is competitive with strong tree-based baselines and is particularly robust under heavy-tailed and heteroskedastic noise.

2 Background and Related Work

Learning to rank. The ranking problem considers two i.i.d. copies (X, Y) and (X', Y'), and constructs a score g leading to a rank rule: $g(X) > g(X') \implies Y > Y'$. The score g can be obtained by minimizing the following pairwise ranking loss (Clémençon et al., 2008):

$$\mathbb{P}\big\{(Y - Y')\,\mathrm{sign}\big(g(X) - g(X')\big) \leq 0\big\}\,, \tag{1}$$

Such objectives are inherently scale-free which is different from regression tasks where the goal is to estimate the conditional mean $m^\star(x) = \mathbb{E}[Y|X = x]$ by minimizing a pointwise loss such as $\mathbb{E}\big[(Y - m(X))^2\big]$.

The learning-to-rank literature, especially in information retrieval, has developed a large array of ranking algorithms to estimate the optimal score g given data $\{(x_i, y_i)\}_{i=1}^n$. Pairwise methods such as RankSVM (Joachims, 2002) and RankNet (Burges et al., 2005) optimize smooth surrogates of pairwise misordering, typically logistic or hinge-style losses applied to score differences $g(x_i) - g(x_j)$, often augmented with heuristic importance weights that reflect ranking metrics like NDCG or MAP (Burges, 2010). Listwise methods such as ListNet, instead construct losses at the level of entire ranked lists, sometimes via differentiable relaxations of the sorting operator (Cao et al., 2007; Taylor et al., 2008). More recent work introduces faster permutation-based relaxations such as *SoftSort*, which approximate ranks and sorted scores with smooth operators suitable for gradient-based training (Prillo & Eisenschlos, 2020).

Recalibration of scores. Many learning algorithms produce scores whose magnitude is not directly interpretable as it is generally not on the same numerical scale as the quantity one ultimately wishes to predict.

Calibration refers to the task of learning a transformation of these raw outputs so that they match a desired target scale. The primary objective of probability calibration is to transform the unnormalized outputs of a classifier into reliable probability estimates that reflect the true empirical likelihood of the target event. Among the most established techniques, Platt scaling assumes a parametric relationship, fitting a sigmoid function to map scores to probabilities, which is particularly effective when the distortion is S-shaped (Platt et al., 1999; Niculescu-Mizil & Caruana, 2005). Alternatively, histogram binning takes a non-parametric approach by partitioning the score space into disjoint intervals and assigning the average label frequency within each bin as the calibrated probability, though this introduces discontinuities (Zadrozny & Elkan, 2001; Naeini et al., 2015). Isotonic regression offers a more flexible non-parametric solution by finding the optimal non-decreasing function that minimizes the squared error between the scores and the targets (Zadrozny & Elkan, 2002). From a theoretical standpoint, this can be interpreted as a shape-constrained regression problem, specifically solving for a univariate monotone projection of the data (often via the Pool Adjacent Violators Algorithm), which allows it to correct arbitrary monotonic distortions without assuming a rigid parametric form (Barlow, 1972; Guntuboyina & Sen, 2018).

Ranking for classification/regression. Combining ranking and calibration methods is an idea that has primarily been applied to classification to improve the accuracy of the class probability estimate. Sculley (2010); Yan et al. (2022) proposed a joint objective approach that employs a single, multi-objective loss function to simultaneously target ranking and calibration. However, this approach forces a trade-off between the ranking and calibration objectives without theoretical guarantee that the calibrated outcome recovers the true class probability. This problem was overcome by Menon et al. (2012), who proposed a sequential approach, learning the ranking first and calibration second. While their methods has theoretical guarantee to recover the true class probability, their analysis was limited to linear scores and binary outcomes, restricting its reach in machine learning applications.

Wüthrich & Ziegel (2024) recently adapted the sequential approach of Menon et al. (2012) to regression with real-valued outcomes. While they analyze the calibration of scores that ostensibly preserve the ordering of the conditional mean, their work does not characterize the class of loss functions guaranteed to produce such scores. Consequently, they do not provide theoretical assurances that the final calibrated estimator converges to the true regression function. Furthermore, their implementation employs the Gamma deviance loss to train first-stage scores, yet the theoretical link between this loss and optimal ranking is unaddressed. They also restrict their analysis to linear scoring models.

Our contributions. In this paper, (i) we propose a two-step regression framework that extends the rank-then-calibrate approach of Menon et al. (2012) to real-valued outcomes. We show that under a specific choice for the ranking loss, this two-stage approach recovers the true regression function at the population level. (ii) We formalize the requirement that we expect of the first stage ranking loss and group them under a single collection of loss functions. We then convert these scores into calibrated predictions using one-dimensional isotonic regression. Our approach provides more flexibility than previous works as it accommodates many ranking loss functions. (iii) Furthermore, while our instantiations of this framework rely on isotonic regression for calibration, we'll show that this choice is not unique and that it may be relaxed to different monotone recalibration mechanisms. (iv) Furthermore, some of our results provide insight on how to interpret the rank losses as we connect them directly to popular measures of associations such as Spearman's ρ and Kendall's τ. (v) Finally, we demonstrate that our method may also be interpreted as a robust alternative to MSE minimization which we show through Numerical Studies.

3 The Decoupled Regression Framework

Consider a two-stage population learning framework to construct a regression model. In Stage 1, a model is trained using a ranking loss $\mathcal{L}_{\text{rank}}$ over a function class $\mathcal{G}$:

$$\text{Stage 1:} \qquad g^\star \in \arg\min_{g \in \mathcal{G}} \mathcal{L}_{\text{rank}}(Y, g(X)). \tag{2}$$

The purpose of Stage 1 is to learn a monotone transformation of the true regression function $m^\star$. Specifically, later, we will show that we expect the optimal model out of Stage 1 to be of the form $g^\star(X) = h(m^\star(X))$ for some h strictly increasing. In Stage 2, Stage 1 output is calibrated with isotonic regression to recover $m^\star$. We apply isotonic regression in Stage 2 which minimizes the squared loss $\mathcal{L}_{\text{iso}}$ over the space of non-decreasing functions $\mathcal{F}$ preserving the ranking learned in Stage 1. Formally,

$$\text{Stage 2:} \qquad f^\star \in \arg\min_{f \in \mathcal{F}} \mathcal{L}_{\text{iso}}(Y, f(g^\star(X))). \tag{3}$$

The isotonic regression step in Stage 2 yields a model $f^\star$ that can be viewed as learning the inverse of the $h(\cdot)$ transformation in $g^\star(X) = h(m^\star(X))$, thereby recovering the true regression function. We will show that $f^\star(g^\star) = m^\star$. Henceforth, we refer to this two-stage framework as the Calibrated After Initial Rank Ordering (CAIRO) framework.

One caveat with Stage 1 optimization is that $g^\star$ itself is only identified up to a strictly increasing transformation. (In classification tasks, this identification issue is absent because there the scale of responses is fixed.) As a consequence, Stage 2 optimizer $f^\star$ is only identified up to (the inverse of) the said transformation as well. However, the composition $f^\star(g^\star) = m^\star$ is clearly always well identified. (If we insist, we could fix Stage 1 identification through requiring $g^\star(X)$ to follow a given distribution, for instance the uniform distribution on $[0, 1]$.) In practice, since in the implementation of Stage 1 we will rely on smoothed versions of the rank loss, Stage 1 identification issue is alleviated. For instance, in Method 1 later based on the log-sigmoid function, Stage 1 optimizer $g^\star$ is identified up to an additive constant, so a unique optimizer can be fixed easily (e.g., by requiring $g^\star(\mathbf{0}) = 0$).

3.1 Stage 1: Optimal-in-Rank-Order Objectives

In this section, we discuss Stage 1 of the decoupled regression framework, and Stage 2 will be discussed in Section 3.2. For two i.i.d. copies (X, Y) and (X', Y') and a symmetric weighting function $w(Y, Y')$, consider a generalization of the pairwise ranking loss in (1):

$$\mathcal{L} \;\equiv\; \mathbb{E}\big[w(Y, Y')\, \mathbb{1}\big\{(Y - Y')\, \operatorname{sign}\big(g(X) - g(X')\big) \le 0\big\}\big]\,. \tag{4}$$

The pairwise structure of (4) aligns with the fundamental objective of learning to rank: to construct a scoring function g that recovers the relative order of the targets. Specifically, the loss uses an indicator function to penalize instances where the scores disagree with the true target order, thereby pushing for the condition $g(X) > g(X') \implies Y > Y'$. Different choices of the weight function $w(Y, Y')$ leads to certain commonly used ranking losses: (i) the *Uniform* weight $w(Y, Y') \equiv 1$ (Clémençon et al., 2008; Burges et al., 2005); (ii) the *Absolute Gap* weight $w(Y, Y') \equiv |Y - Y'|$ (Agarwal & Niyogi, 2005; Wang et al., 2018); and (iii) the *Rank Gap* weight $w(Y, Y') \equiv |F_Y(Y) - F_Y(Y')|$ (Menon & Williamson, 2016). The introduction of these weights allows for a flexible ranking objective.

Additional insights can be obtained from Theorem 1 below, which provides the equivalent pointwise forms of loss (4) with the specified weighting functions. This gives the theoretical basis for using the pointwise ranking loss in Stage 1 in (2). The proof of Theorem 1 is provided in Appendix B.3

Theorem 1. *Let $\mathcal{L}$ be the loss defined in (4). Then we have the following equivalent representations $\mathcal{L}_{\text{uni}}$, $\mathcal{L}_{\text{abs}}$, and $\mathcal{L}_{\text{cdf}}$ based on three different choices of the weight $w(Y, Y')$:*

1. ***Kendall's τ:*** *If $w(Y, Y') = 1$, minimizing $\mathcal{L}$ with respect to g is equivalent to minimizing $\mathcal{L}_{\text{uni}}$ involving Kendall's rank correlation coefficient $\tau \equiv \mathbb{E}[\text{sign}(Y - Y')\ \text{sign}(g(X) - g(X'))]$ between Y and $g(X)$:*

$$\mathcal{L}_{\text{uni}}(Y, g(X)) = \frac{1}{2}(1 - \tau(Y, g(X))). \tag{5}$$

2. ***Gini Covariance:*** *If $w(Y, Y') = |Y - Y'|$, minimizing $\mathcal{L}$ with respect to g is equivalent to minimizing $\mathcal{L}_{\text{abs}}$ involving the Gini Covariance between Y and the rank of $g(X)$:*

$$\mathcal{L}_{\text{abs}}(Y, g(X)) = c - 2\,\text{Cov}(Y, F_{g(X)}(g(X))). \tag{6}$$

   Here $c = \frac{1}{2}\mathbb{E}|Y - Y'|$, this term does not affect optimization of $\mathcal{L}_{\text{abs}}$ with respect to g.

3. ***Spearman's ρ:*** *If $w(Y, Y') = |F_Y(Y) - F_Y(Y')|$, minimizing $\mathcal{L}$ with respect to g is equivalent to minimizing $\mathcal{L}_{\text{cdf}}$ involving Spearman's rank correlation $\rho_S \equiv 12 \cdot \text{Cov}(F_Y(Y), F_g(g(X)))$ between Y and $g(X)$:*

$$\mathcal{L}_{\text{cdf}}(Y, g(X)) = \frac{1}{6}(1 - \rho_S(Y, g(X))). \tag{7}$$

Theorem 1 provides substantial interpretability for these weighted ranking losses by connecting them to classical statistical measures. (i) For $w = 1$, as established in Theorem 1, this loss acts as a direct proxy for Kendall's τ. Thus, penalizing every misordered pair equally is mathematically equivalent to maximizing the probability of concordance between the target Y and the model score $g(X)$. (ii) For $w = |Y - Y'|$, Theorem 1 reveals that this loss is the negative of the Gini covariance (up to a constant). It follows that assigning higher penalties to pairs distant in the target space is equivalent to maximizing the association between the response Y and the score's cumulative distribution $F_{g(X)}(g(X))$. The Gini Covariance is a standard tool in actuarial science for quantifying the relationship between risk scores and insurance losses and is extensively studied by Yitzhaki & Schechtman (2012). (iii) For $w = |F_Y(Y) - F_Y(Y')|$, Theorem 1 confirms that the loss is a linear transformation of Spearman's rank correlation ρ_S: hence, prioritizing pairs that are far apart in quantile space is equivalent to maximizing the correlation between the ranks of Y and $g(X)$. This variant offers a compromise; it captures the relative distribution of the targets while remaining invariant to extreme outliers in Y that might otherwise destabilize the Gini objective.

We now characterize the family of ranking losses suitable for the first stage of the CAIRO framework. We focus specifically on the subclass of losses that recover the correct ordering of the conditional mean as their optimal solution. To formalize this requirement, we define the following property:

Definition 1. *A loss function $\mathcal{L}$ is Optimal in Rank Order (ORO) if its population minimizer belongs to $\mathcal{G}'$ which is defined as:*

$$\mathcal{G}' \equiv \left\{h(m^\star) : \ h : \mathbb{R} \to \mathbb{R} \textit{ strictly increasing}\right\} \subset \mathcal{G}.$$

This definition is central to the CAIRO framework because the success of calibration in Stage 2 depends entirely on the ordering established in the ranking stage. Since calibration applies monotonic transformation, it can only adjust the scale of predictions while strictly preserving their order. Consequently, if the Stage 1 objective fails to satisfy the ORO property, the resulting scores $g^\star$ will not align with the ranking of the conditional mean $m^\star$; in such cases, no amount of recalibration can recover the true regression function. By restricting our attention to ORO losses, we ensure that the decoupled framework remains theoretically consistent with standard regression objectives.

The following theorem shows that the three losses defined in Theorem 1 achieve the ORO property under specific conditions:

Theorem 2. *Let (X, Y) and (X', Y') be i.i.d., and define $m^\star(x) \equiv \mathbb{E}[Y|X = x]$. The weighted pairwise ranking losses defined in Theorem 1 are ORO under the following conditions:*

1. ***Gini Covariance:*** *Provided $\mathbb{E}[|Y|] < \infty$, $\mathcal{L}_{\text{abs}}$ is ORO.*
2. ***Kendall's τ:*** *$\mathcal{L}_{\text{uni}}$ is ORO if the distribution of Y is pairwise consistent with $m^\star$:*

$$m^\star(x) > m^\star(x') \quad \iff \quad \mathbb{P}(Y > Y'|X = x,\, X' = x') > \frac{1}{2}.$$

3. ***Spearman's ρ:*** *$\mathcal{L}_{\text{cdf}}$ is ORO if the Additive-Noise Model holds:*

$$Y = m^\star(X) + \varepsilon,$$

where $\varepsilon \perp X$ and ε has a continuous, strictly increasing CDF.

Remark. *The results in Theorem 2 illustrate how different ranking objectives rely on varying structural assumptions to achieve the ORO property. (i) As can be seen in the proof of Theorem 2 in Appendix B.4, the Gini Covariance objective is governed directly by the difference in conditional means $m^\star(x) - m^\star(x')$; consequently, it satisfies the ORO property under very mild assumptions. (ii) The Kendall objective targets the probability of concordance, requiring only that the likelihood of $Y > Y'$ remains consistent with the ordering of the means. (iii) In contrast, the Spearman objective operates in quantile space; because the ordering of expected quantiles does not always coincide with the ordering of expectations, this objective requires the stronger structural assumption of additive noise to guarantee the ORO property.*

3.2 Stage 2: Calibration via Isotonic Regression

In Stage 2 we restore the scale of the regression function. Let $m^\star(x) = \mathbb{E}[Y|X = x]$ and let $g^\star$ denote Stage 1 score. Under any ORO loss, an optimal population ranker satisfies $g^\star(x) = h(m^\star(x))$ for some strictly increasing $h : \mathbb{R} \to \mathbb{R}$. The goal of Stage 2 is to learn a monotone map that inverts the unknown transform h and recovers $m^\star$.

The calibration problem seeks a nondecreasing function that best aligns these scores with the outcome Y. Let $\mathcal{F} \equiv \{f : \mathbb{R} \to \mathbb{R} \text{ nondecreasing}\}$. We define the population version of the objective:

$$f^\star \in \arg\min_{f\in\mathcal{F}} \mathbb{E}\big[(Y - f(g^\star(X)))^2\big]. \tag{8}$$

When composed with $g^\star$, the optimizer of (8) recovers $m^\star$, so calibration recovers the correct regression function. The following theorem formalizes this statement; the proof is given in Appendix B.1.

Theorem 3. *Assume $\mathbb{E}[Y^2] < \infty$ and suppose Stage 1 score satisfies $g^\star(x) = h(m^\star(x))$ for some strictly increasing $h : \mathbb{R} \to \mathbb{R}$. In Stage 2, let $f^\star$ be any minimizer of the monotone least-squares problem in* (8)*. Then it recovers the true regression function*

$$f^\star\big(g^\star(X)\big) = m^\star(X).$$

4 Estimation and Algorithms

We implement the CAIRO framework by approximating the population losses from Section 3 using an i.i.d. sample $\{(x_i, y_i)\}_{i=1}^n$. Since these objectives are non-differentiable, we derive empirical analogs in both pairwise and pointwise forms and apply smooth surrogates to enable gradient-based optimization. The second stage solves the empirical isotonic regression problem. This section formalizes these smoothing strategies and analyzes the resulting computational complexity.

Pairwise empirical losses (Stage 1). We first consider the empirical version of the pairwise losses for Stage 1. Assuming no ties in the targets y_i's, and symmetric weights $w_{ij} = w_{ji} \geq 0$, we define the empirical version of (4):

$$\ell(y, g(x)) \equiv \frac{1}{n(n-1)} \sum_{1\leq i<j\leq n} w_{ij}\, \mathbb{1}\big\{(y_i - y_j)\big(g(x_i) - g(x_j)\big) \leq 0\big\}\,. \tag{9}$$

This template unifies our Stage 1 objectives at the empirical level through the choice of w_{ij}. Specifically, we utilize three weights: (i) the uniform weights $w_{\text{uni}} \equiv 1$, (ii) the absolute gap weights $w_{\text{abs}} \equiv |y_i - y_j|$, and (iii) the rank gap weights $w_{\text{cdf}} \equiv |\hat{F}_n(y_i) - \hat{F}_n(y_j)|$, where $\hat{F}_n$ is the empirical CDF of Y. For computational purposes, we consider an equivalent form of (9) by isolating the misordered pairs (see proof in Appendix B.5)

$$\ell(y, g(x)) \;=\; \frac{1}{n(n-1)} \sum_{i\neq j} w_{ij}\, \mathbb{1}\{y_i > y_j\}\, \mathbb{1}\{g(x_i) < g(x_j)\}, \tag{10}$$

The hard indicator function in (10) results in a non-differentiable objective, we replace it with a smooth surrogate. Specifically, observe that the indicator function respects the following upper bound:

$$\mathbb{1}\{g(x_i) < g(x_j)\} \ \leq \ \log_2\Big(1 + e^{-\sigma\,(g(x_i) - g(x_j))}\Big)\,, \tag{11}$$

We refer to the right hand side of (11) as the log-sigmoid function (that is evaluated at $g(x_i) - g(x_j)$), where σ is a hyperparameter that characterizes the shape of the sigmoid. The log-sigmoid function has been extensively used as a smooth, convex surrogate for the indicator function in recent literature since an upper bound on (10) can be derived using (11):

$$\ell(y, g(x)) \ \leq \ \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij} \mathbb{1}\{y_i > y_j\} \ \log_2\Big(1 + e^{-\sigma\,(g(x_i) - g(x_j))}\Big)\,. \tag{12}$$

In practice, we would use the right hand side of (12) as the loss function in the implementation. As an upper bound it naturally decreases the true empirical loss function (10).

We can see that the RankNet loss is (12) by setting $w_{ij} = w_{\text{uni}} = 1$. This result shows that the RankNet loss can be viewed either as cross-entropy between the true probability and a sigmoid-modeled posterior probability (Burges et al., 2005), or as a convex upper bound on the pairwise error, corresponding to two alternative interpretations of RankNet: either a probabilistic pairwise classifier or a smooth surrogate for improving Kendall's τ, respectively. This fact supports our ORO property by showing that it includes standard, widely used ranking losses.

Pointwise empirical losses (Stage 1). Next we consider the empirical version of the pointwise losses in Theorem 1. We construct these losses by replacing the population CDF with its sample counterpart, which can then be written using the *rank* function (55). Because the *rank* operator is piecewise constant with vanishing gradients, we employ a differentiable *softrank* approximation (Blondel et al., 2020; Cuturi et al., 2019) to enable gradient-based optimization. We illustrate this approach with the Gini Covariance loss $\mathcal{L}_{\text{abs}}$. Consider the empirical version: $\ell_{\text{abs}}(y, g(x)) = -\operatorname{Cov}_n\big(y_i, \widehat{F}_{g(X)}(g(x_i))\big)$. We show in Appendix B.6 that ℓ_{abs} may be written equivalently, up to positive scalars and added constants as:

$$\ell_{\text{abs}}(y, g(x)) \ \equiv \ -\sum_{i=1}^{n} y_i \ \operatorname{rank}\big(g(x_i)\big). \tag{13}$$

We can smoothen (13) by replacing the *rank* function with *softrank* (Blondel et al., 2020; Cuturi et al., 2019), yielding the smooth objective $\ell_{\text{soft-abs}}$ defined as:

$$\ell_{\text{soft-abs}}(y, g(x)) \equiv \ -\sum_{i=1}^{n} y_i \ \operatorname{softrank}\big(g(x_i)\big), \tag{14}$$

Remark. (13) *has been studied by Frees et al. (2011) and Yitzhaki & Schechtman (2012) as a metric for evaluating insurance risk scores and measures of association.. Wang et al. (2018) used* (13) *to derive* (10) *with $w_{ij} = |y_i - y_j|$ which they implemented with log-sigmoid smoothing. To our knowledge, this work is the* *first to analyze* (14) *as a differentiable pointwise loss function implemented via smooth rank approximations.*

Empirical Calibration (Stage 2). Having learned a scoring rule $\hat{g}(x_i)$ from a Stage 1 loss for data $\{(x_i, y_i)\}_{i=1}^n$, we now restore the target scale via monotone calibration by solving the empirical analogue of (8) to the dataset $\{(\hat{g}(x_i), y_i)\}_{i=1}^n$:

$$\min_f \frac{1}{n}\sum_{i=1}^n \big(y_i - f(\hat{g}(x_i))\big)^2 \quad \text{s.t.} \quad f(\hat{g}(x_i)) \le f(\hat{g}(x_j)) \text{ whenever } \hat{g}(x_i) \le \hat{g}(x_j). \tag{15}$$

Let $\hat{f}$ denote the resulting isotonic fit defined at the observed scores $\hat{g}(x_i)$. To ensure a continuous mapping, we utilize the implementation from Pedregosa et al. (2011), which linearly interpolates between the fitted steps and clips predictions outside the training range. For prediction we then use

$$\hat{m}(x) \equiv \hat{f}\big(\hat{g}(x)\big),$$

Computational considerations. The computational complexity of CAIRO is governed by the choice of empirical loss in Stage 1. When optimizing pairwise objectives (12), we employ log-sigmoid smoothing, which requires computing differences across all pairs in a batch, yielding $O(n^2)$ complexity. Conversely, for pointwise objectives (13), we utilize the differentiable *softrank* operator, which Blondel et al. (2020) demonstrated can be computed in $O(n \log n)$ time. Since Stage 2 calibration via the Pool Adjacent Violators (PAV) algorithm requires only linear time on sorted data (Barlow, 1972), the overall training complexity scales as $O(n^2)$ for pairwise losses and $O(n \log n)$ for pointwise losses.

5 Numerical Studies

We empirically validate the CAIRO framework through controlled simulations and real-world regression benchmarks. Our primary objective is to assess regression and ranking accuracy across varying noise regimes, isolating the impact of the Stage 1 ranking objective.

CAIRO model variants. We implement three model variants of the CAIRO framework. All variants share the same Stage 2 procedure (isotonic calibration) but differ in their Stage 1 loss and smoothing strategy:

- `RankNet`: RankNet loss (12) with uniform weights $w_{ij} = 1$ and log-sigmoid smoothing.
- `RankNet-GiniW`: RankNet loss (12) with absolute value weights $w_{ij} = |y_i - y_j|$ and log-sigmoid smoothing.
- `GiniNet-SoftRank`: Pointwise Gini Covariance loss (14) with *softrank* smoothing.

Stage 1 scores are parameterized by a two-layer MLP (32×16, ReLU), optimized via Adam.

Baselines and metrics. We benchmark against three regression standards: (i) `NN-MSE`, an MLP with the identical architecture trained on squared error; (ii) `RandomForest` (RF), an ensemble of CART trees; and (iii) `LightGBM`, a gradient-boosted decision tree model minimizing MSE. Performance is evaluated using Root Mean Squared Error (RMSE) for regression accuracy, and Spearman's ρ_S and Kendall's τ for ranking quality.

5.1 Synthetic Data-Generating Processes

To isolate the effect of noise distribution, we fix the underlying signal to be linear. Covariates $X \in \mathbb{R}^d$ and weights w are drawn from $\mathcal{N}(0, I_d)$, with $n = 6000$ and $d = 10$. We define the linear function $\eta = X^\top w/\sqrt{d}$ and generate responses Y by applying three distinct mechanisms to this linear function η:

1. *Normal:* A standard linear regression setting where $Y = \eta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$. This serves as a baseline where MSE is theoretically optimal.
2. *Gamma:* We apply a canonical log-link, setting $\mu = \exp(\eta)$. Responses are drawn from $Y \sim \mathrm{Gamma}(k, \mu/k)$ with shape $k = 2$. Here, variance scales with the mean, introducing heteroskedasticity.
3. *Heteroskedastic Tail:* We again use the mean $\mu = \exp(\eta)$ but introduce multiplicative heavy-tailed noise: $Y = \mu + \varepsilon \cdot \sqrt{\mu}$, where ε is drawn from a Lognormal distribution.

5.2 Results on Synthetic Data

Performance under homoskedastic noise. In the *Normal* setting, the `NN-MSE` baseline is theoretically optimal; it achieves the lowest RMSE as expected. `GiniNet-SoftRank` proves highly effective here, matching the baseline's precision while maintaining strong ranking metrics. `RankNet-GiniW` underperforms slightly, suggesting that pointwise objectives (via SoftRank) are better suited in high-SNR regimes.

Robustness to heavy-tailed noise. The advantages of the CAIRO framework emerge in the *Heteroskedastic Tail* setting. Here, standard MSE minimization fails, as the squared error loss is disproportionately sensitive to extreme outliers. In contrast, all CAIRO variants remain stable. Notably, `RankNet-GiniW` achieves the best performance, outperforming both the baseline and the `GiniNet-SoftRank`. We attribute this robustness to the log-sigmoid component of the pairwise loss, which saturates for large error magnitudes, effectively capping the gradient contribution of outliers. While `GiniNet-SoftRank` still improves upon the MSE baseline, it does not match the robustness of the pairwise approach in this regime.

5.3 Real-Data Analysis

To assess generalization, we evaluate our method on five regression benchmarks from the UCI Machine Learning Repository (Frank & Asuncion, 2010): Auto MPG (AUTO), Communities and Crime (CCRI), Computer Hardware (COMP), Concrete Slump Test (CSLU), and Abalone (ABALONE). These datasets vary in size ($n \in [103, 4177]$) and dimensionality ($p \in [6, 102]$); see Table 2.

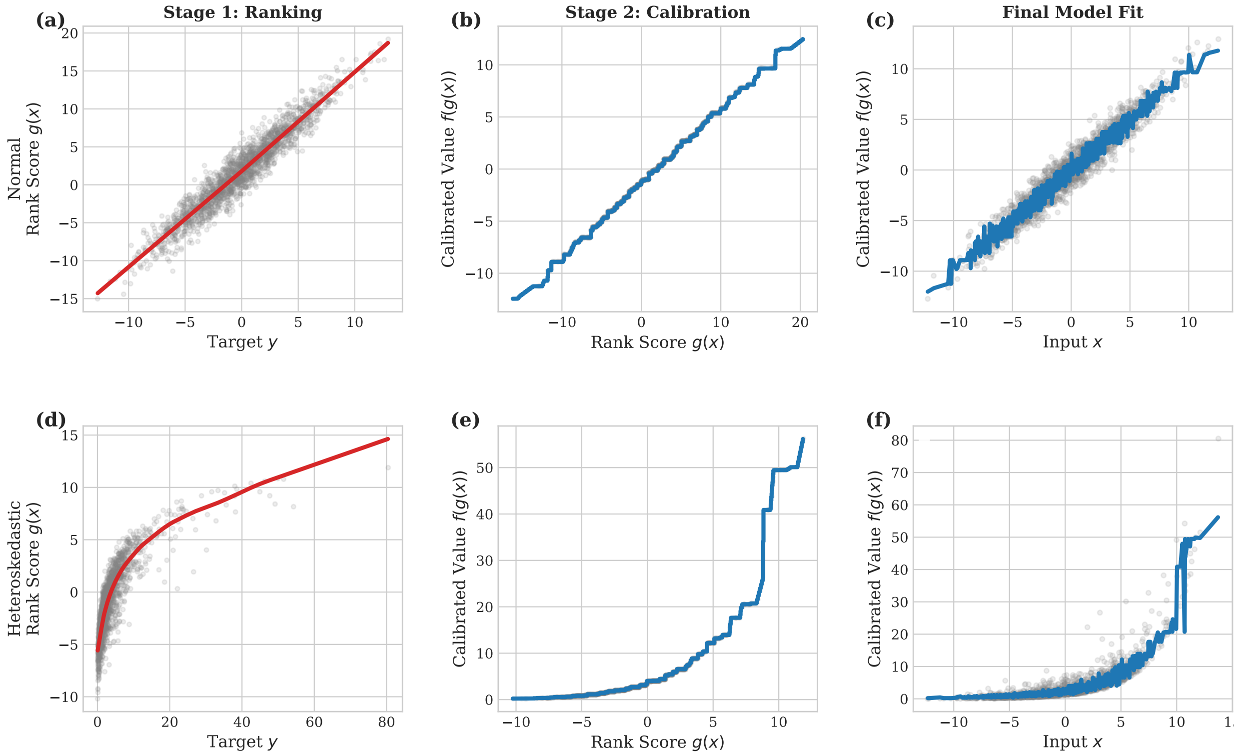

Figure 1: `CAIRO-RankNet` across Data Regimes. **Top Row (Normal Regime):**Stage 1 (a) learns a linear ranking, and Stage 2 (b) learns a linear calibration map, resulting in a standard regression fit (c). **Bottom Row (Heteroskedastic Regime):** In the presence of heavy-tailed noise, Stage 1 (d) learns a non-linear ranking to preserve order. Stage 2 (e) learns a non-parametric step function to correct this warp. The final model (f) successfully recovers the underlying signal (blue line) despite the large variance in the raw data (gray cloud).

Preprocessing. We follow the preprocessing protocol of Roy & Larocque (2012). For AUTO, we remove the high-cardinality *car name* feature and incomplete cases ($n = 392$). For CCRI, we discard predictors with $> 58\%$ missing values ($p = 102$). For ABALONE, we regress strictly on continuous shell measurements to predict Age. All datasets are devoid of missing values after preprocessing.

Results. We compare CAIRO variants against the regression baselines detailed in Section 5. Table 1 reports mean performance across five random splits.

First, we observe that `NN-MSE` exhibits significant instability, particularly on smaller datasets like AUTO and CSLU. In contrast, the CAIRO variants, specifically `RankNet-GiniW` and `GiniNet-SoftRank`, demonstrate much better results using identical architectures. This suggests that the rank-based objectives in CAIRO variants may stabilize neural networks in low-data regimes.

Second, CAIRO proves competitive with tree ensembles, which are typically dominant in tabular domains. On the *Computer Hardware* (COMP) dataset, characterized by significant price outliers, `RankNet-GiniW` achieves an RMSE of 73.7, substantially outperforming Random Forest (87.8) and LightGBM (94.2); this mirrors our synthetic results regarding the robustness of pairwise losses. On *Concrete Slump* (CSLU), the smallest dataset ($n = 103$), CAIRO outperforms all baselines (RMSE 3.52 vs. 4.45 for RF), indicating superior data efficiency. Finally, on the larger ABALONE task, CAIRO yields the highest ranking correlations ($\rho \approx 0.78$) and lowest RMSE (2.12). These results demonstrate that CAIRO effectively closes the performance gap between neural networks and tree-based models on tabular regression tasks.

Second, CAIRO variants remain competitive with tree-based models, which typically dominate tabular regression. On the *Computer Hardware* (COMP) dataset, which contains significant price outliers, the `RankNet-GiniW` variant achieves an RMSE of 73.7, outperforming both `RandomForest` (87.8) and `LightGBM` (94.2). This performance aligns with our synthetic findings. Finally, on the larger ABALONE task, `RankNet` yields the highest ranking correlations ($\rho \approx 0.78$) and lowest RMSE (2.12). These results demonstrate that the proposed framework allows neural networks to match or exceed the performance of state-of-the-art tree models on varied tabular benchmarks.

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A Appendix

Table 1: Real-data performance on five UCI regression tasks. Entries are mean (95% CI) over 5 random 70/30 train–test splits. Best value per dataset/metric is in bold.

Dataset	Model	Spearman	Kendall's τ	RMSE
AUTO (Auto MPG)				
	CAIRO–GiniNet-SoftRank	0.870 ± 0.022	0.719 ± 0.022	4.244 ± 0.294
	CAIRO–RankNet	0.933 ± 0.011	0.795 ± 0.020	3.188 ± 0.204
	CAIRO–RankNet-GiniW	0.923 ± 0.025	0.787 ± 0.027	3.405 ± 0.455
	NN-MSE	0.050 ± 0.578	0.026 ± 0.434	12.321 ± 4.540
	Random Forest	$\mathbf{0.937 \pm 0.015}$	$\mathbf{0.798 \pm 0.017}$	$\mathbf{2.994 \pm 0.239}$
	LightGBM–MSE	0.935 ± 0.015	0.796 ± 0.019	3.010 ± 0.230
CCRI (Communities and Crime)				
	CAIRO–GiniNet-SoftRank	0.810 ± 0.018	$\mathbf{0.640 \pm 0.020}$	0.142 ± 0.002
	CAIRO–RankNet	0.757 ± 0.013	0.586 ± 0.013	0.160 ± 0.005
	CAIRO–RankNet-GiniW	0.762 ± 0.010	0.593 ± 0.011	0.159 ± 0.004
	NN-MSE	0.475 ± 0.371	0.358 ± 0.280	0.190 ± 0.053
	Random Forest	$\mathbf{0.817 \pm 0.015}$	0.631 ± 0.016	$\mathbf{0.138 \pm 0.004}$
	LightGBM–MSE	0.812 ± 0.012	0.626 ± 0.013	$\mathbf{0.138 \pm 0.003}$
COMP (Computer Hardware)				
	CAIRO–GiniNet-SoftRank	0.784 ± 0.042	0.644 ± 0.042	105.055 ± 64.484
	CAIRO–RankNet	0.800 ± 0.072	0.657 ± 0.070	89.822 ± 31.436
	CAIRO–RankNet-GiniW	0.833 ± 0.043	0.688 ± 0.045	$\mathbf{73.741 \pm 34.310}$
	NN-MSE	0.392 ± 0.363	0.297 ± 0.294	334.020 ± 142.277
	Random Forest	$\mathbf{0.904 \pm 0.022}$	$\mathbf{0.753 \pm 0.031}$	87.826 ± 27.136
	LightGBM–MSE	0.808 ± 0.077	0.638 ± 0.079	94.157 ± 27.191
CSLU (Concrete Slump Test)				
	CAIRO–GiniNet-SoftRank	0.769 ± 0.114	0.604 ± 0.106	5.207 ± 0.895
	CAIRO–RankNet	0.822 ± 0.189	0.717 ± 0.145	4.807 ± 2.306
	CAIRO–RankNet-GiniW	$\mathbf{0.900 \pm 0.013}$	$\mathbf{0.760 \pm 0.027}$	$\mathbf{3.522 \pm 0.420}$
	NN-MSE	0.033 ± 0.409	0.038 ± 0.320	23.394 ± 12.779
	Random Forest	0.890 ± 0.030	0.733 ± 0.043	4.445 ± 0.394
	LightGBM–MSE	0.844 ± 0.034	0.682 ± 0.042	4.519 ± 0.251
ABALONE				
	CAIRO–GiniNet-SoftRank	0.631 ± 0.040	0.503 ± 0.034	2.610 ± 0.042
	CAIRO–RankNet	$\mathbf{0.781 \pm 0.014}$	$\mathbf{0.637 \pm 0.013}$	$\mathbf{2.115 \pm 0.042}$
	CAIRO–RankNet-GiniW	0.778 ± 0.015	0.635 ± 0.013	2.116 ± 0.036
	NN-MSE	0.634 ± 0.028	0.481 ± 0.023	3.255 ± 0.102
	Random Forest	0.763 ± 0.019	0.606 ± 0.017	2.172 ± 0.023
	LightGBM–MSE	0.761 ± 0.015	0.603 ± 0.014	2.192 ± 0.045

Table 2: Characteristics of the real data sets used in our study.

Data set	No. of data points (n)	No. of predictors (p)	No. of unique response values	Source
AUTO	392	7	127	UCI
CCRI	1994	102	98	UCI
COMP	209	6	116	UCI
CSLU	103	7	90	UCI
ABALONE	4177	7	29	UCI

Table 3: Synthetic-data performance across noise models: mean (95% CI) over 5 repetitions. Best value per scenario/metric is in bold.

Scenario	Model	RMSE	Spearman	Kendall's τ
Normal ($\sigma = 1.0$)				
	CAIRO–RankNet	1.093 ± 0.026	0.594 ± 0.074	0.435 ± 0.059
	CAIRO–RankNet-GiniW	1.095 ± 0.025	0.588 ± 0.074	0.431 ± 0.056
	CAIRO–GiniNet-SoftRank	1.031 ± 0.010	0.654 ± 0.051	$\mathbf{0.484 \pm 0.041}$
	NN-MSE	$\mathbf{1.004 \pm 0.008}$	$\mathbf{0.667 \pm 0.045}$	0.481 ± 0.037
	Random Forest	1.039 ± 0.020	0.642 ± 0.045	0.459 ± 0.037
	LightGBM–MSE	1.033 ± 0.019	0.643 ± 0.056	0.461 ± 0.046
Gamma (shape $k = 2$)				
	CAIRO–RankNet	2.238 ± 0.726	0.701 ± 0.058	0.527 ± 0.050
	CAIRO–RankNet-GiniW	2.082 ± 0.577	0.714 ± 0.058	0.540 ± 0.050
	CAIRO–GiniNet-SoftRank	$\mathbf{2.040 \pm 0.554}$	0.730 ± 0.040	$\mathbf{0.553 \pm 0.037}$
	NN-MSE	2.125 ± 0.775	$\mathbf{0.736 \pm 0.040}$	0.544 ± 0.037
	Random Forest	2.176 ± 0.783	0.698 ± 0.034	0.510 ± 0.030
	LightGBM–MSE	2.112 ± 0.687	0.695 ± 0.037	0.508 ± 0.032
Heteroskedastic tail				
	CAIRO–RankNet	$\mathbf{1.423 \pm 0.089}$	0.858 ± 0.037	0.693 ± 0.043
	CAIRO–RankNet-GiniW	1.449 ± 0.095	0.860 ± 0.039	0.696 ± 0.044
	CAIRO–GiniNet-SoftRank	1.580 ± 0.239	$\mathbf{0.863 \pm 0.033}$	$\mathbf{0.698 \pm 0.039}$
	NN-MSE	1.882 ± 0.513	0.848 ± 0.044	0.667 ± 0.047
	Random Forest	1.873 ± 0.533	0.830 ± 0.026	0.642 ± 0.028
	LightGBM–MSE	1.705 ± 0.470	0.854 ± 0.026	0.672 ± 0.030

B Results and Proofs

For clarity of exposition, our theoretical analysis assumes that the score distribution is continuous, i.e., $\mathbb{P}\big(g(X) = g(X')\big) = 0$. However, this assumption is not strictly necessary. In the presence of ties, our results regarding the Gini covariance and Spearman's correlation remain valid provided that the standard cumulative distribution function is replaced by the *mid-distribution function* (Parzen, 2004). This function is defined as $\widetilde{F}_g(t) = \mathbb{P}\big(g(X) < t\big) + \frac{1}{2}\,\mathbb{P}\big(g(X) = t\big)$. Unlike the standard CDF, the mid-distribution ensures that the centered expectation property $\mathbb{E}\big[\widetilde{F}_g(g(X))\big] = 1/2$ holds even when $\mathbb{P}\big(g(X) = t\big) > 0$, thereby preserving the symmetry required for the covariance identities derived in Theorem 1.

B.1 Proof of Composition Theorem (Theorem 3)

Proof. Let g be the output from the first stage. Then $g = h(m^\star)$ for a strictly increasing function h. We want to characterize the second-stage function f such that

$$f = \arg\min_{t \in \mathcal{M}} \mathbb{E}\left[(Y - t(g(X)))^2\right].$$

By adding and subtracting $m^\star$ in the objective and expanding the square, we obtain

$$\begin{aligned}
\mathbb{E}\big[(Y - t(g(X)))^2\big] &= \mathbb{E}\big[(Y - m^\star(X) + m^\star(X) - t(g(X)))^2\big] \\
&= \mathbb{E}\big[(Y - m^\star(X))^2\big] + \mathbb{E}\big[(m^\star(X) - t(g(X)))^2\big] \\
&\quad + 2\,\mathbb{E}[(Y - m^\star(X))\,(m^\star(X) - t(g(X)))].
\end{aligned}$$

The first term $\mathbb{E}\big[(Y - m^\star(X))^2\big]$ is independent of t, since it depends only on the true regression function $m^\star$. We apply the law of iterated expectations:

$$\begin{aligned}
\mathbb{E}_{Y,X}[(Y - m^\star(X))\,(m^\star(X) - t(g(X)))] &= \mathbb{E}_X\left[\mathbb{E}_{Y|X}[(Y - m^\star(X))\,(m^\star(X) - t(g(X)))\mid X]\right] \\
&= \mathbb{E}_X\left[(m^\star(X) - t(g(X)))\,\mathbb{E}_{Y|X}[Y - m^\star(X)\mid X]\right] \\
&= 0,
\end{aligned}$$

since $\mathbb{E}_{Y|X}[Y - m^\star(X) \mid X] = 0$. Hence the objective simplifies to

$$\arg\min_{t \in \mathcal{M}} \mathbb{E}\left[(Y - t(g(X)))^2\right] = \arg\min_{t \in \mathcal{M}} \mathbb{E}\left[(m^\star(X) - t(g(X)))^2\right].$$

This is minimized when $t(g(X)) = m^\star(X)$, i.e.,

$$t(h(m^\star(X))) = m^\star(X) \quad \Rightarrow \quad t = h^{-1}.$$

The inverse exists since h is strictly increasing, and $h^{-1} \in \mathcal{M}$ because it is non-decreasing as the inverse of a strictly increasing function. Therefore, the optimal second-stage function is $f^\star = h^{-1}$, and

$$f^\star(g(X)) = h^{-1}(h(m^\star(X))) = m^\star(X).$$ □

B.2 Useful Results

Lemma 1. *Let (X, Y) and (X', Y') be i.i.d. with finite second moments, then*

$$\mathbb{E}\big[(Y - Y')(X - X')\big] \;=\; 2\,\mathrm{Cov}(Y, X).$$

Proof.

$$\begin{aligned}
\mathbb{E}\big[(Y-Y')(X-X')\big] &= \mathbb{E}[YX] - \mathbb{E}[YX'] - \mathbb{E}[Y'X] + \mathbb{E}[Y'X'] \\
&= \mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X'] - \mathbb{E}[Y']\mathbb{E}[X] + \mathbb{E}[Y'X'] \\
&= \mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[YX] \\
&= 2\big(\mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X]\big) = 2\,\mathrm{Cov}(Y,X).
\end{aligned}$$

In the second line, independence across pairs gives $\mathbb{E}[YX'] = \mathbb{E}[Y]\mathbb{E}[X']$ and $\mathbb{E}[Y'X] = \mathbb{E}[Y']\mathbb{E}[X]$. In the third line, i.i.d. across pairs yields $\mathbb{E}[Y'X'] = \mathbb{E}[YX]$, $\mathbb{E}[X'] = \mathbb{E}[X]$, and $\mathbb{E}[Y'] = \mathbb{E}[Y]$. □

Lemma 2. *Let X and X' be i.i.d. real random variables with continuous cdf F. With $\mathrm{sgn}(t) = \mathbb{1}\{t > 0\} - \mathbb{1}\{t < 0\}$, we have*

$$\mathbb{E}\big[\mathrm{sgn}(X-X') \mid X\big] = 2F(X) - 1 \quad \textit{a.s.}$$

Proof. Starting with the decomposition $\mathrm{sgn}(X-X') = \mathbb{1}\{X > X'\} - \mathbb{1}\{X < X'\}$, we have

$$\begin{aligned}
\mathbb{E}[\mathrm{sgn}(X-X')|X] &= \mathbb{E}[\mathbb{1}\{X > X'\} - \mathbb{1}\{X < X'\}|X] \\
&= \mathbb{P}(X > X'|X) - \mathbb{P}(X < X'|X) \\
&= F(X) - (1 - F(X)) \\
&= 2F(X) - 1.
\end{aligned}$$

□

Proposition 1. *Let (X,Y) and (X',Y') be i.i.d. with finite second moments and independent across pairs. Then*

$$\mathrm{Cov}\big(Y,\, F_{g(X)}(g(X))\big) = \frac{1}{2}\,\mathbb{E}\big[(Y-Y')\,\mathbb{1}\{\,g(X) > g(X')\,\}\big]\,.$$

Proof. Let (X',Y') be an i.i.d. copy of (X,Y). Then

$$\begin{aligned}
2\cdot\mathrm{Cov}\big(Y, F_{g(X)}(g(X))\big) &= 2\Big(\mathbb{E}\big[Y\,F_{g(X)}(g(X))\big] - \mathbb{E}[Y]\,\mathbb{E}\big[F_{g(X)}(g(X))\big]\Big) \\
&= 2\Big(\mathbb{E}\big[Y\,F_{g(X)}(g(X))\big] - \tfrac{1}{2}\,\mathbb{E}[Y]\Big) \\
&= 2\,\mathbb{E}\big[Y\,F_{g(X)}(g(X))\big] - \mathbb{E}[Y] \\
&= 2\,\mathbb{E}\Big[Y\,\mathbb{E}_{X'|Y,X}\big[\mathbb{1}\{g(X') \le g(X)\} \mid Y, X\big]\Big] - \mathbb{E}[Y] \\
&= 2\,\mathbb{E}\big[Y\,\mathbb{1}\{g(X') \le g(X)\}\big] - \mathbb{E}[Y].
\end{aligned} \tag{16}$$

where we have use independence of Y and X' to condition on Y in the inner expectation in line 4. We also applied the law of total expectation between the pair (Y,X) and X' to obtain the last line. Now, by symmetry of the i.i.d. pairs (X,Y) and (X',Y'),

$$2\,\mathbb{E}\big[Y\,\mathbb{1}\{g(X') \le g(X)\}\big] = \mathbb{E}\big[Y\,\mathbb{1}\{g(X') \le g(X)\}\big] + \mathbb{E}\big[Y'\,\mathbb{1}\{g(X) \le g(X')\}\big]. \tag{17}$$

We may replace $\leq$ by $<$ in the indicators. Combining (16) and (17), we obtain

$$\begin{aligned}
2 \cdot \mathrm{Cov}\big(Y, F_{g(X)}(g(X))\big) &= \mathbb{E}\big[Y\, \mathbb{1}\{g(X') < g(X)\}\big] + \mathbb{E}\big[Y'\, \mathbb{1}\{g(X) < g(X')\}\big] - \mathbb{E}[Y] \\
&= \mathbb{E}\big[Y\, \mathbb{1}\{g(X') < g(X)\}\big] + \mathbb{E}\big[Y'\, \mathbb{1}\{g(X) < g(X')\}\big] \\
&\quad - \mathbb{E}\big[Y\big(\mathbb{1}\{g(X') < g(X)\} + \mathbb{1}\{g(X) < g(X')\}\big)\big] \\
&= \mathbb{E}\big[Y'\, \mathbb{1}\{g(X) < g(X')\}\big] - \mathbb{E}\big[Y\, \mathbb{1}\{g(X) < g(X')\}\big] \\
&= \mathbb{E}\big[(Y' - Y)\, \mathbb{1}\{g(X) < g(X')\}\big] \\
&= \mathbb{E}\big[(Y - Y')\, \mathbb{1}\{g(X) > g(X')\}\big]. && (18)
\end{aligned}$$

This proves

$$2 \cdot \mathrm{Cov}\big(Y, F_{g(X)}(g(X))\big) = \mathbb{E}\big[(Y - Y')\, \mathbb{1}\{g(X) > g(X')\}\big],$$

which is the desired identity. ☐

B.3 Proof of Theorem 1

Proof. For any nonnegative symmetric weight $w(Y, Y')$ with $\mathbb{E}[w(Y, Y')] < \infty$, define

$$\begin{aligned}
\mathcal{L}(g) &\equiv \mathbb{E}\big[w(Y, Y')\, \mathbb{1}\big\{(Y - Y')\, \mathrm{sign}\big(g(X) - g(X')\big) \leq 0\big\}\big]\,. && (19) \\
&= \mathbb{E}\big[w(Y, Y')\, \mathbb{1}\big\{(Y - Y')\big(g(X) - g(X')\big) < 0\big\}\big]\,. && (20)
\end{aligned}$$

1) Uniform weight and Kendall's τ. Take $w(Y, Y') \equiv 1$, then $\mathcal{L} \to \mathcal{L}_{\text{uni}}$. Also apply the following transformation:

$$\mathbb{1}\big\{(Y - Y')\big(g(X) - g(X')\big) < 0\big\} = \frac{1 - \mathrm{sign}(Y - Y')\, \mathrm{sign}\big(g(X) - g(X')\big)}{2} \tag{21}$$

Plugging this into (20) yields

$$\mathcal{L}_{\text{uni}}(g) = \frac{1}{2} - \frac{1}{2}\, \mathbb{E}\Big[\, \mathrm{sign}(Y - Y')\, \mathrm{sign}\big(g(X) - g(X')\big)\Big].$$

Under (A1), the population Kendall correlation between Y and $g(X)$ is

$$\begin{aligned}
\tau\big(Y, g(X)\big) &\equiv \mathbb{E}\Big[\, \mathrm{sign}(Y - Y')\, \mathrm{sign}\big(g(X) - g(X')\big)\Big]. && (22) \\
&= 1 - 2 \cdot \mathcal{L}_{\text{uni}}(g) && (23)
\end{aligned}$$

2) Absolute gap weight and the Gini covariance. Take $w(Y, Y') = |Y - Y'|$, then $\mathcal{L} \to \mathcal{L}_{\text{abs}}$.

$$\mathcal{L}_{\text{abs}}(g) = \mathbb{E}\big[\,|Y - Y'|\,\mathbb{1}\big\{(Y - Y')\big(g(X) - g(X')\big) < 0\big\}\big] \tag{24}$$

$$= \mathbb{E}\left[\frac{|Y - Y'|}{2} - \frac{|Y - Y'|\ \text{sign}(Y - Y')\ \text{sign}\big(g(X) - g(X')\big)}{2}\right]. \tag{25}$$

$$= \frac{1}{2}\,\mathbb{E}|Y - Y'| - \frac{1}{2}\,\mathbb{E}\Big[(Y - Y')\ \text{sign}\big(g(X) - g(X')\big)\Big]. \tag{26}$$

where we used the transformation of (21) in the second line and also that $|Y - Y'|\,\text{sign}(Y - Y') = Y - Y'$ to go from second to third equality. Now write $\text{sign}\big(g(X) - g(X')\big) = \mathbb{1}\{g(X) > g(X')\} - \mathbb{1}\{g(X) < g(X')\}$ Therefore, by plugging this in (26) and further manipulating by swapping the i.i.d. pairs $(X, Y) \leftrightarrow (X', Y')$ and using symmetry,

$$\mathbb{E}\Big[(Y - Y')\ \text{sign}\big(g(X) - g(X')\big)\Big] = \mathbb{E}\Big[(Y - Y')\,\mathbb{1}\{g(X) > g(X')\}\Big] - \mathbb{E}\Big[(Y - Y')\,\mathbb{1}\{g(X) < g(X')\}\Big]. \tag{27}$$

$$= 2\,\mathbb{E}\Big[(Y - Y')\,\mathbb{1}\{g(X) > g(X')\}\Big]. \tag{28}$$

Invoking Proposition 1, we combine it with (28) to get

$$\mathbb{E}\Big[(Y - Y')\ \text{sign}\big(g(X) - g(X')\big)\Big] = 4\,\text{Cov}(Y,\ F_g(g(X)))\,.$$

Plugging into (26) yields the claimed identity:

$$\mathcal{L}_{\text{abs}}(g) = \frac{1}{2}\,\mathbb{E}|Y - Y'| - 2\,\text{Cov}(Y,\ F_g(g(X)))\,,$$

3) CDF-rank gap weight and Spearman's ρ_S. Take $w(Y, Y') = \big|F_Y(Y) - F_Y(Y')\big|$, then $\mathcal{L} \to \mathcal{L}_{\text{cdf}}$ where F_Y is the CDF of Y. Under the assumption, F_Y is continuous and strictly increasing on the support of Y, so $\text{sign}(Y - Y') = \text{sign}\big(F_Y(Y) - F_Y(Y')\big)$. Consequently,

$$\mathcal{L}_{\text{cdf}}(g) = \mathbb{E}\big[\,|F_Y(Y) - F_Y(Y')|\,\mathbb{1}\big\{(Y - Y')\big(g(X) - g(X')\big) < 0\big\}\big] \tag{29}$$

$$= \mathbb{E}\big[\,|F_Y(Y) - F_Y(Y')|\,\mathbb{1}\big\{\text{sign}(Y - Y')\,\text{sign}(g(X) - g(X')) < 0\big\}\big] \tag{30}$$

$$= \mathbb{E}\big[\big|F_Y(Y) - F_Y(Y')\big|\,\mathbb{1}\big\{\text{sign}\big(F_Y(Y) - F_Y(Y')\big) \neq \text{sign}\big(g(X) - g(X')\big)\big\}\big]$$

$$= \frac{1}{2}\,\mathbb{E}\big|F_Y(Y) - F_Y(Y')\big| - \frac{1}{2}\,\mathbb{E}\Big[\big(F_Y(Y) - F_Y(Y')\big)\ \text{sign}\big(g(X) - g(X')\big)\Big]. \tag{31}$$

First term. $F_Y(Y)$ is $\text{Unif}(0, 1)$, so

$$\mathbb{E}\big|F_Y(Y) - F_Y(Y')\big| = \int_0^1\!\!\int_0^1 |u - v|\,du\,dv = \frac{1}{3}.$$

Second term. Apply the sign-to-indicator identity (28) with Y replaced by $F_Y(Y)$, and then apply

Proposition 1 again, now with the random variable $F_Y(Y)$ in place of Y:

$$\mathbb{E}\Big[\big(F_Y(Y) - F_Y(Y')\big)\ \mathrm{sign}\big(g(X) - g(X')\big)\Big] = 2\,\mathbb{E}\Big[\big(F_Y(Y) - F_Y(Y')\big)\ \mathbb{1}\{g(X) > g(X')\}\Big]. \tag{32}$$
$$= 4\mathrm{Cov}(F_Y(Y),\ F_g(g(X))) \tag{33}$$

Substituting into (31) yields

$$\mathcal{L}_{\mathrm{cdf}}(g) = \frac{1}{2}\cdot\frac{1}{3} - 2\,\mathrm{Cov}(F_Y(Y),\ F_g(g(X))) = \frac{1}{6} - 2\,\mathrm{Cov}(F_Y(Y),\ F_g(g(X))) \equiv \frac{1}{6} - \frac{\rho_S(Y, g(X))}{6}$$

This completes the proof of all three claims. □

B.4 Proof of Theorem 2: ORO property

Proof. Fix (x, x') and consider the conditional pairwise ranking loss

$$\mathcal{L}_w(Y, g(X); x, x') \ \equiv\ \mathbb{E}\big[w(Y,Y')\,\mathbb{1}\big\{(Y - Y')\big(g(x) - g(x')\big) < 0\big\}\ \big|\ X = x,\ X' = x'\big]\,,$$

where the inequality is strict under (A1) (no ties in Y). The optimal choice is to pick the sign of $g(x) - g(x')$ that yields the smaller conditional risk; i.e. one should choose $g(x) > g(x')$ whenever

$$\mathbb{E}\big[w(Y,Y')\,\mathbb{1}\{Y < Y'\}\,\big|\,X = x, X' = x'\big] \le \mathbb{E}\big[w(Y,Y')\,\mathbb{1}\{Y > Y'\}\,\big|\,X = x, X' = x'\big] \tag{34}$$
$$\implies \quad \mathbb{E}\big[w(Y,Y')\big(\mathbb{1}\{Y < Y'\} - \mathbb{1}\{Y > Y'\}\big)\,\big|\,X = x, X' = x'\big] \le 0 \tag{35}$$
$$\implies \quad \mathbb{E}\big[w(Y,Y')\ \mathrm{sign}(Y - Y')\,\big|\,X = x, X' = x'\big] \ge 0\,. \tag{36}$$

If we define the quantity $S_w(x,x') \equiv \mathbb{E}[w(Y,Y')\,\mathrm{sign}(Y - Y')|\ X = x, X' = x']$, then the pairwise optimal decision rule is $\mathrm{sign}\big(g(x) - g(x')\big) = \mathrm{sign}\big(S_w(x,x')\big)$ Consequently, to prove the ORO property it suffices to show that, for each of the three weights,

$$\mathrm{sign}\big(S_w(x,x')\big) = \mathrm{sign}\big(m^\star(x) - m^\star(x')\big) \tag{37}$$

Indeed, if (37) holds, then any population minimizer must order pairs (x, x') the same way as $m^\star$, which implies $g^\star = h(m^\star)$ for some strictly increasing h, i.e. $g^\star \in \mathcal{G}'$. Conversely, any strictly increasing transform of $m^\star$ induces the same ordering and therefore achieves the same minimal conditional risk for all pairs (x, x'), yielding global optimality.

Case 1: $w(y, y') \equiv 1$. Here:

$$S_{\mathrm{uni}}(x,x') = \mathbb{E}\big[\mathrm{sign}(Y - Y') \mid X = x, X' = x'\big] \tag{38}$$
$$= \mathbb{P}(Y > Y' \mid x, x') - \mathbb{P}(Y < Y' \mid x, x') \tag{39}$$
$$= \mathbb{P}(Y > Y' \mid x, x') - (1 - \mathbb{P}(Y > Y' \mid x, x')) \tag{40}$$
$$= 2\,\mathbb{P}(Y > Y' \mid x, x') - 1. \tag{41}$$

where we have assumed assumption no ties in the third equation to sum both probabilities to 1. By assumption, $\mathbb{P}(Y > Y' \mid x, x') > \frac{1}{2}$ if and only if $m^\star(x) > m^\star(x')$, which gives (37) for $w \equiv 1$.
Case 2: $w(y, y') = |y - y'|$. Here:

$$\begin{aligned} S_{\text{abs}}(x, x') &= \mathbb{E}\big[\,|Y - Y'|\,\text{sign}(Y - Y') \mid X = x, X' = x'\big] && (42)\\ &= \mathbb{E}\big[\,Y - Y' \mid X = x, X' = x'\big] && (43)\\ &= \mathbb{E}[Y \mid X = x] - \mathbb{E}[Y' \mid X' = x'] && (44)\\ &= m^\star(x) - m^\star(x'). && (45)\end{aligned}$$

In the second line we used the identity $|t|\,\text{sign}(t) = t$. Therefore

$$\text{sign}\big(S_{\text{abs}}(x, x')\big) = \text{sign}\big(m^\star(x) - m^\star(x')\big),$$

so (37) holds for the absolute-gap weight.

Case 3: $w(y, y') = |F_Y(y) - F_Y(y')|$. Under Spearman's ρ in case 3, F_Y is continuous and strictly increasing, hence $\text{sign}\big(F_Y(Y) - F_Y(Y')\big) = \text{sign}(Y - Y')$, therefore:

$$\begin{aligned} S_{\text{cdf}}(x, x') &= \mathbb{E}\big[\,|F_Y(Y) - F_Y(Y')|\,\text{sign}(Y - Y') \mid X = x, X' = x'\big] && (46)\\ &= \mathbb{E}\big[\,|F_Y(Y) - F_Y(Y')|\,\text{sign}\big(F_Y(Y) - F_Y(Y')\big) \mid X = x, X' = x'\big] && (47)\\ &= \mathbb{E}\big[\,F_Y(Y) - F_Y(Y') \mid X = x, X' = x'\big] && (48)\\ &= \mathbb{E}[F_Y(Y) \mid X = x] - \mathbb{E}\big[F_Y(Y') \mid X' = x'\big] && (49)\\ &= \eta(x) - \eta(x'), && (50)\end{aligned}$$

where $\eta(x) \equiv \mathbb{E}[F_Y(Y) \mid X = x]$. In the third line we used the identity $|t|\,\text{sign}(t) = t$. It follows that

$$\text{sign}\big(S_{\text{cdf}}(x, x')\big) = \text{sign}\big(\eta(x) - \eta(x')\big),$$

so the minimizers of $\mathcal{L}_{\text{cdf}}$ are exactly the strictly increasing transforms of η. Under the (sufficient) part of the assumption where $Y = m^\star(X) + \varepsilon$ with ε independent of X and with strictly increasing CDF, the family of conditional distributions $\{Y \mid X = x\}$ is strictly stochastically ordered by $m^\star(x)$. Since F_Y is strictly increasing, the map $x \mapsto \eta(x) = \mathbb{E}[F_Y(Y) \mid X = x]$ is strictly increasing in $m^\star(x)$, i.e. there exists a strictly increasing h such that $\eta = h(m^\star)$. Consequently,

$$\text{sign}\big(S_{\text{cdf}}(x, x')\big) = \text{sign}\big(m^\star(x) - m^\star(x')\big),$$

and (37) holds for the rank-gap weight as well. This completes the proof of all three claims. $\square$

B.5 Alternative Expression for the Empirical Weighted ranking loss

Proposition 2. *Assume* $\{(x_i, y_i)\}_{i=1}^n$ *is such that there are no ties in* $\{y_i\}$ *and in* $\{g(x_i)\}$*, and* $w_{ij} = w_{ji} \geq 0$ *for all* $i \neq j$*. Then the empirical loss defined in (9) satisfies*

$$\ell(g) \;=\; \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij}\, \mathbb{1}\{y_i > y_j\}\, \mathbb{1}\{g(x_i) < g(x_j)\}.$$

Proof. The event $(y_i - y_j)\big(g(x_i) - g(x_j)\big) < 0$ occurs if and only if exactly one of the two differences is reversed:

$$\mathbb{1}\big\{(y_i - y_j)\big(g(x_i) - g(x_j)\big) < 0\big\} = \mathbb{1}\{y_i > y_j,\, g(x_i) < g(x_j)\} + \mathbb{1}\{y_i < y_j,\, g(x_i) > g(x_j)\}.$$

Plugging this into (9) yields

$$\ell(g) = \frac{1}{n(n-1)} \sum_{1 \leq i < j \leq n} w_{ij} \Big[\mathbb{1}\{y_i > y_j,\, g(x_i) < g(x_j)\} + \mathbb{1}\{y_i < y_j,\, g(x_i) > g(x_j)\}\Big]. \tag{51}$$

Now consider the quantity S defined as:

$$\begin{aligned} S &\equiv \sum_{i \neq j} w_{ij}\, \mathbb{1}\{y_i > y_j\}\, \mathbb{1}\{g(x_i) < g(x_j)\}. &(52)\\ &= \sum_{1 \leq i < j \leq n} \Big[w_{ij}\, \mathbb{1}\{y_i > y_j,\, g(x_i) < g(x_j)\} + w_{ji}\, \mathbb{1}\{y_j > y_i,\, g(x_j) < g(x_i)\}\Big] &(53)\\ &= \sum_{1 \leq i < j \leq n} w_{ij} \Big[\mathbb{1}\{y_i > y_j,\, g(x_i) < g(x_j)\} + \mathbb{1}\{y_i < y_j,\, g(x_i) > g(x_j)\}\Big]. &(54) \end{aligned}$$

where we rewrote S by grouping terms over unordered pairs $\{i, j\}$ with $i < j$ in the second line and used the fact that $y_j > y_i$ and $g(x_j) < g(x_i)$ are exactly the negations of $y_i > y_j$ and $g(x_i) < g(x_j)$ in the third. Comparing this with the expression for $\ell(g)$ above, we obtain

$$\ell(g) = \frac{1}{n(n-1)}\, S = \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij}\, \mathbb{1}\{y_i > y_j\}\, \mathbb{1}\{g(x_i) < g(x_j)\},$$

which is exactly the claimed identity. $\square$

B.6 Rank based expression of the empirical Gini covariance loss

Proof. We derive the rank-based expression for the empirical Gini-covariance objective used in Stage 1. Let $\mathrm{Cov}_n(\cdot,\cdot)$ denote the empirical covariance, $\mathrm{Cov}_n(A_i, B_i) \equiv \frac{1}{n}\sum_{i=1}^n (A_i - \bar{A})(B_i - \bar{B})$, and define the empirical CDF of the scores by $\widehat{F}_{g(X)}(t) \equiv \frac{1}{n}\sum_{j=1}^n \mathbb{1}\{g(x_j) \le t\}$. Note that:

$$\widehat{F}_{g(X)}(g(x_i)) = \frac{1}{n}\sum_{j=1}^n \mathbb{1}\{g(x_j) \le g(x_i)\} = \frac{\mathrm{rank}(g(x_i))}{n}. \tag{55}$$

Plugging (55) into the loss gives

$$\begin{aligned}
\ell_{\text{abs}}(y, g(x)) &\equiv -2\,\mathrm{Cov}_n\Big(y_i, \frac{\mathrm{rank}(g(x_i))}{n}\Big) \\
&= -\frac{2}{n}\,\mathrm{Cov}_n\big(y_i, \mathrm{rank}(g(x_i))\big) \\
&= -\frac{2}{n}\cdot\frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})\Big(\mathrm{rank}(g(x_i)) - \overline{\mathrm{rank}}\Big),
\end{aligned} \tag{56}$$

where $\overline{\mathrm{rank}} = \frac{1}{n}\sum_{i=1}^n \mathrm{rank}(g(x_i))$. Since $\{\mathrm{rank}(g(x_i))\}_{i=1}^n$ is a permutation of $\{1,\ldots,n\}$, we have $\overline{\mathrm{rank}} = \frac{1}{n}\sum_{r=1}^n r = \frac{n+1}{2}$. Expanding (56) and using $\sum_{i=1}^n (y_i - \bar{y}) = 0$ yields

$$\begin{aligned}
\ell_{\text{abs}}(y, g(x)) &= -\frac{2}{n^2}\sum_{i=1}^n (y_i - \bar{y})\Big(\mathrm{rank}(g(x_i)) - \tfrac{n+1}{2}\Big) \\
&= -\frac{2}{n^2}\sum_{i=1}^n (y_i - \bar{y})\,\mathrm{rank}(g(x_i)) + \frac{2}{n^2}\cdot\frac{n+1}{2}\sum_{i=1}^n (y_i - \bar{y}) \\
&= -\frac{2}{n^2}\sum_{i=1}^n (y_i - \bar{y})\,\mathrm{rank}(g(x_i)) \\
&= -\frac{2}{n^2}\left(\sum_{i=1}^n y_i\,\mathrm{rank}(g(x_i)) - \bar{y}\sum_{i=1}^n \mathrm{rank}(g(x_i))\right) \\
&= -\frac{2}{n^2}\left(\sum_{i=1}^n y_i\,\mathrm{rank}(g(x_i)) - \bar{y}\cdot\frac{n(n+1)}{2}\right).
\end{aligned} \tag{57}$$

The second term in (57) is a constant with respect to g, therefore, up to an additive constant and a positive scalar, minimizing ℓ_{abs} is equivalent to minimizing

$$-\sum_{i=1}^n y_i\,\mathrm{rank}(g(x_i)),$$

which is the desired rank-based form. $\square$